Pranava Karthekeyan Ponvinayagan

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Objective

Electrical engineering student with hands-on research experience in closed-loop neurostimulation, real-time embedded systems, and brain-computer interfaces. Seeking a research or engineering internship starting in May 2027, applying real-time signal processing, hardware-software integration, and firmware development to advance neural engineering and biomedical technology.

Education

Georgia Institute of Technology | Atlanta, GA August 2026 – Present
B.S. Electrical Engineering (Circuit Technology & Bioengineering) Expected Graduation, December 2028

Skills

Languages: C++, Python
Hardware & Instrumentation: Brain Products ActiChamp (EEG), Magstim BiStim² (TMS), Delsys Trigno (EMG), ESP32-S3, T-SIM7600G-H (Cellular/GPS), MAX30105 (SpO2/HR)
Software & Tools: RTXI, PlatformIO, Git, Xenomai RTOS, TCP/IP, Serial/UART, LSL, Cloud Backend Development
Industry Concepts: Brain-Computer Interfaces (BCI), Closed-Loop Control, Neural Engineering, Neurostimulation, Real-Time Systems, Biosignal Processing, Embedded Systems, Firmware, IoT, Real-Time Data Acquisition, Geofencing, GPS Tracking

Experience

Movement Neuroscience Lab @ Northeastern University | Boston, MA *June 2026* – August *2026*
Visiting Researcher

- *EEG-Modulated Neurostimulation Platform for Force Tracking:* Engineered and led a full-stack, closed-loop neurostimulation platform (C++/RTXI, real-time Linux) unifying multi-modal brain stimulation, EEG acquisition, and visuomotor force-tracking on one synchronized timeline, letting a subject trace in real time the waveform being delivered to their own brain
    - Architected 5 RTXI modules including 6-input force collector/processor, 32-channel EEG collector (using LSL)/processor with real-time spectral feature extraction from Brain Products ActiChamp Amplifier, a unified stimulation engine driving 8 configurable channels (tES, dual TI channels, a sum-of-sines channel, & TMS triggering), a fully designable per-trial task generation, and a physics-calibrated subject-facing force display
    - Integrated real-time closed-loop control modulation for stimulation amplitude from live, baseline-calibrated EEG band power with slew-rate limiting for safety; added alpha-frequency/phase extraction toward frequency- and phase-locked neuromodulation
    - Developed a Python orchestrator that automates full experiment deployment, including RTXI environment build and launch, parameter configuration for RTXI modules, live waveform preview, a preset system, unified hardware I/O routing, and experimental control, turning a manual multi-step setup into a single reproducible workflow for lab usage

Movement Neuroscience Lab @ Northeastern University | Boston, MA *April 2026 – May 2026*
Research Intern

- *Neural Acquisition and Stimulation System:* Developed C++ RTXI plugins on Xenomai real-time Linux, replacing a non-real-time setup with programmatic TMS control and multichannel EMG acquisition for live human MEP experiments
    - Implemented and hardware-validated a serial communication plugin for the Magstim BiStim² paired-pulse TMS device, driven by a state-machine architecture with a checksummed command protocol, enabling real-time, closed-loop-ready stimulation control confirmed via loopback testing
    - Constructed a 16-channel EMG acquisition module for the Delsys Trigno system using a dual-socket TCP architecture and a wait-free non-RT listener thread, enabling lag-free real-time data streaming into the RTXI pipeline

Projects

CareWatch: Dementia Patient Monitoring System | Franklin, MA *January 2026 – March 2026*
Independent Developer

- Engineered T-SIM7600G-H (ESP32-S3) cellular wearable firmware as the backbone of a caregiver-facing remote monitoring system for dementia patients, with non-blocking polling, self-registration device pairing, and a 5-button elderly-friendly navigation GUI
- Programmed cellular and GPS connectivity for real-time location tracking using adaptive geofencing across three caregiver-defined security zones, and a MAX30105 pulse oximetry sensor for SpO2, heart rate, and breathing rate monitoring, plus an SOS trigger system, turning raw sensor data into actionable safety alerts for caregivers